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Multimodal Techniques for Financial Time-Series Forecasting

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Abstract

Accurate stock price forecasting requires integrating heterogeneous signals ranging from raw price histories to unstructured financial news and structured corporate filings. Recent multimodal datasets such as FinMultiTime enable cross-modal financial analysis, but the comparative performance of different model architectures on such data remains underexplored. We present a systematic benchmarking study on S&P 500 constituents using three complementary modalities: historical price time series, financial news sentiment, and structured fundamental features from SEC XBRL filings. We compare unimodal baselines (LSTM, PatchTST, TSMixer, TCN-Transformer, TFT) against pretrained foundation models (Chronos T5-Large, Chronos-2), multimodal fusion architectures (TFT-FinBERT, TCN-FinBERT, ChronosFinBert, FinBERT+CNN+Transformer), and graph neural networks (GATv2, EvolveGCN). Our central finding is that the choice of fusion mechanism dominates the choice of temporal encoder: cross-attention and encoder-decoder fusion between a temporal encoder and FinBERT improve on unimodal baselines, while concatenation-based late fusion underperforms them due to modality imbalance in the shared bottleneck. Graph neural networks offer complementary cross-sectional structure that sequence models cannot represent. These findings provide practical guidance for designing multimodal financial forecasting systems at scale.

Contents

1 Introduction **3**

2 Problem Statement **3**

3 Related Work **4**

4 Solution and Methodology **6**

4.1 Dataset 6

4.1.1 Stock Price Data 6

4.1.2 Financial News Data 6

4.1.3 Financial Reports & Earnings Data 7

4.2 Financial Reports & Earnings Data 7

4.3 Sector Relational Data 8

4.4 Model Architectures 8

4.4.1 Unimodal Temporal Baselines 8

4.4.2 Temporal Encoder for Financial Time-Series (Chronos-T5 + FinBERT Fusion) 11

4.4.3 Multimodal Fusion Architectures (TFT/TCN + FinBERT) 12

4.4.4 FinBERT + CNN Price Encoder + Transformer Decoder 13

4.4.5 Graph Neural Network Architectures 13

4.5 Fusion Strategies 14

5 Discussion **15**

6 Results **16**

7 Conclusion **18**

1 Introduction

Financial time-series forecasting is a longstanding challenge at the intersection of economics, statistics, and machine learning. Traditional approaches rely exclusively on historical price data that often fail to capture the abrupt, sentiment-driven dynamics of real-world financial markets.

At the same time, financial decision-making inherently depends on heterogeneous information sources, including news, corporate disclosures, and market signals. Recent advances in deep learning and large language models (LLMs) have enabled the extraction of meaningful signals from unstructured data such as financial news, while structured sources like SEC filings provide reliable indicators of firm fundamentals. This shift reflects a broader trend toward multimodal learning, where integrating diverse data sources improves predictive performance.

Datasets such as FinMultiTime have further accelerated this direction by aligning multiple modalities — including text, tables, and time series — into unified forecasting frameworks. However, while such datasets enable multimodal modeling, they also introduce new challenges in understanding how different data sources and model architectures interact.

2 Problem Statement

This project investigates the following research question: *How effectively can multimodal deep learning architectures combine historical price time series, financial news sentiment, and structured financial features to improve short-horizon stock price forecasting on S&P 500 equities?*

Formally, let $\mathbf{x} = \{x_{t-T+1}, \ldots, x_t\}$ denote a lookback window of historical close prices for a given stock, where T is the input sequence length and x_t is the closing price at trading day t. In addition to this price sequence, each sample may include aligned financial news text and periodic financial statement data representing sentiment-driven and fundamental information, respectively.

The forecasting objective is to predict the next H trading-day close prices,

$$\mathbf{y} = \{x_{t+1}, \ldots, x_{t+H}\},$$

where H is the prediction horizon. Across the experiments described in this report, $H \in \{5, 7\}$ trading days and $T \in \{20, 30, 60, 90\}$ trading days, depending on the architectural variant under evaluation. Most experiments use $H = 5$; the multimodal FinBERT–TCN configuration uses $H = 7$. The graph models use $T = 20$, the baseline temporal models use $T \in \{30, 60\}$, and ChronosFinBert uses $T \in \{30, 60, 90\}$ in its ablation.

The central challenge is therefore not only to model temporal dependencies in stock prices, but also to determine whether heterogeneous modalities can be fused in a way that improves predictive accuracy relative to unimodal baselines. This framing allows us to compare how different architectures exploit complementary information from market behavior, event-driven news signals, and lower-frequency financial disclosures under a consistent short-horizon forecasting task.

3 Related Work

Financial time-series forecasting has developed across four methodological strands that this study brings into a common benchmark.

Classical and recurrent sequence models. ARIMA and GARCH-family models [4] provided the earliest rigorous treatment of price and volatility dynamics but assumed linear conditional structure. The shift to deep learning began with the LSTM [5], whose gated memory cell made it the default recurrent baseline for sequence forecasting for over a decade. Temporal convolutional networks [8] demonstrated that dilated causal convolutions could match or exceed recurrent networks on long sequences at lower computational cost.

Transformer and MLP-based forecasters. Attention-based architectures [23] were adapted to time series by PatchTST [6], which segments each channel into patches and processes them independently through a Transformer encoder with channel-independent RevIN normalization. TSMixer [7] moved in the opposite direction, showing that an all-MLP architecture alternating time-mixing and feature-mixing blocks could match Transformer accuracy at a fraction of the parameter count. The Temporal Fusion Transformer [20] combined variable selection, recurrent encoding, and interpretable multi-head attention into a unified architecture aimed at multi-horizon forecasting with built-in feature importance.

Pretrained time-series foundation models. Chronos [2] adapted the T5 encoder-decoder framework [24] to tokenized time series, enabling zero-shot probabilistic forecasts on unseen series. Chronos-2 [9] extended this direction with improved tokenization and group attention. Their applicability to financial returns, where the distributional assumptions underlying time-series pretraining may not hold, is an open empirical question that this work addresses.

Financial NLP and text-conditioned forecasting. Tetlock [11] established empirically that media sentiment carries short-horizon predictive information for equity returns. FinBERT [1] adapted BERT [10] to financial corpora and has since been used less as a classifier and more as a frozen or lightly tuned feature extractor inside larger forecasting pipelines. Baltrušaitis et al. [13] frame the core problems of multimodal learning — representation, alignment, fusion, translation — that recur in any text-plus-price system.

Multimodal financial datasets. Wenyan [18] and the related FinMultiTime release [19] provide aligned multimodal inputs over S&P 500 and HS300 constituents, enabling controlled comparison of fusion architectures. Our work uses the S&P 500 subset of this release.

Graph neural networks for equities. GAT [14] and GATv2 [15] attend over neighboring nodes with learned weights, which is well-suited to cross-sectional equity modeling where sector

peers and correlated stocks transmit information. EvolveGCN [16] extends graph convolution to dynamic graphs by evolving the convolution parameters over time, and heterogeneous temporal graph networks [17] accommodate multiple node and edge types. These approaches model inter-stock dependencies that univariate sequence models ignore by construction.

Our study draws on all four strands and evaluates how much each contributes when placed inside a consistent short-horizon forecasting benchmark.

4 Solution and Methodology

4.1 Dataset

All experiments in this project are conducted on the S&P 500 Multimodal Financial Dataset [18], a large-scale benchmark covering multiple modalities of information for time-series forecasting. The original dataset integrates four modalities — financial news text, financial reports, technical chart images, and stock price time series — across both U.S. (S&P 500) and Chinese (HS300) markets.

In this work, we construct a reduced and task-specific version of the dataset by excluding image data and HS300 (Chinese market) samples, focusing solely on structured and textual signals from the U.S. market.

Dataset Type	**Size**	**Records**	**Frequency Count**	**Frequency**
Semi-Annual Table	84.04 GB	2,676	8,028	Semi-Annual
Quarterly/Annual Timeseries	1.83 GB	4,213	4,213	Quarterly / Annual
Daily Text	14.1 GB	4,694	3,351,852	Daily
Total (All)	108.64 GB	—	3,559,440	—

Table 1: Summary of the reduced S&P 500 multimodal dataset used in this study.

4.1.1 Stock Price Data

Daily stock price data including open, high, low, close, and volume (OHLCV). These series are normalized per stock to ensure stationarity and comparability across assets.

4.1.2 Financial News Data

Financial news articles associated with each stock, including headlines and full text. Articles are summarized and processed to extract sentiment signals, providing qualitative context for price movements.

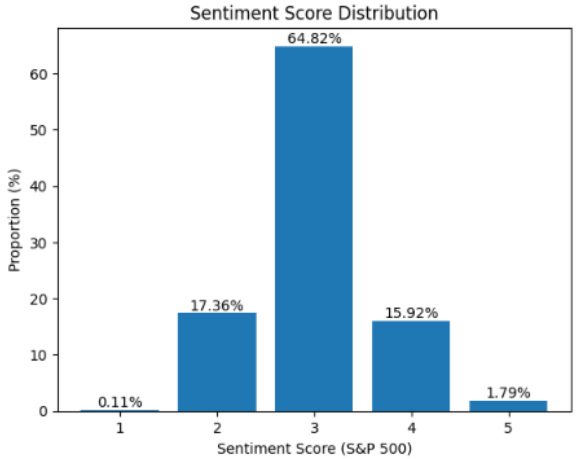

Figure 1: Distribution of article-level sentiment labels in the news subset.

For the S&P 500 subset, sentiment labels were represented on a five-point ordinal scale: 1 denoted negative, 2 denoted somewhat negative, 3 denoted neutral, 4 denoted somewhat positive, and 5 denoted positive. This encoding preserved polarity strength while remaining compatible with downstream aggregation and multimodal fusion.

4.1.3 Financial Reports & Earnings Data

Structured financial indicators were derived from company filings, including key variables related to profitability, liquidity, and capital structure, such as cash flow, equity, and earnings. These variables were aligned with reporting periods and forward-filled to match trading days so that fundamental signals could be synchronized with daily market observations.

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SEC Fundamental Data Preprocessing. Structured financial indicators were derived from company filings, covering profitability, liquidity, and capital structure (e.g., cash flow, equity, earnings). These were aligned with reporting periods and forward-filled to match trading days, synchronizing fundamental signals with daily observations.

4.3 Sector Relational Data

Each ticker was associated with its GICS sector and used to construct sector-based relational edges in graph models, enabling the pipeline to represent shared industry exposure and sector-level co-movement.

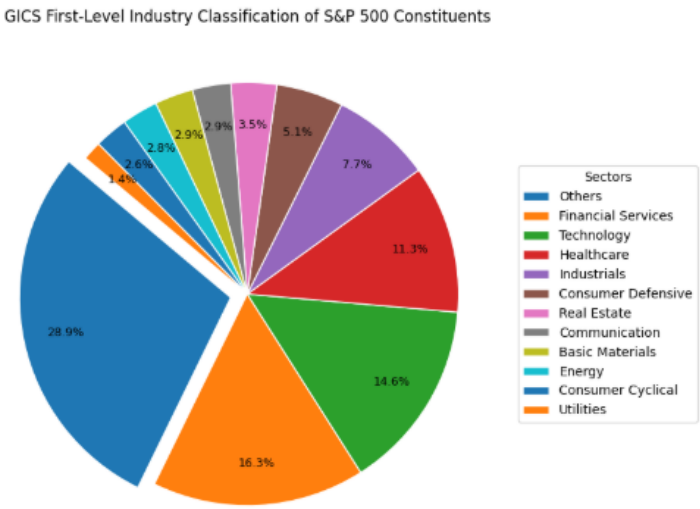

Figure 2: Distribution of stocks across GICS sectors in the relational metadata.

4.4 Model Architectures

4.4.1 Unimodal Temporal Baselines

We evaluate six families of unimodal temporal encoders, trained on price-derived feature sequences with lookback and target specifications that vary by model (see the individual paragraphs below). A seventh recurrent baseline (GRU) was implemented with a 60-day lookback over six engineered features but only validated on a single-ticker sample, and is therefore not reported in the aggregate comparison.

LSTM. A two-layer stacked LSTM with hidden dimensions 128 and 64, trained on 26 engineered OHLCV-derived features. The model uses a direct multi-step output head that predicts all five future log returns in a single forward pass, avoiding compounding autoregressive error. A composite loss combining Huber regression [22] with an optional directional binary cross-entropy term encourages both calibrated magnitude and sign prediction.

The 26-feature input set combines log returns, rolling return statistics at 5-, 10-, and 20-day windows, RSI(14), ATR(14), MACD and MACD signal, Bollinger bandwidth over 20 days, lagged returns at lags 1, 2, 3, and 5, and intraday range structure variables (high-low range, close-open gap, upper

and lower shadow, VWAP proxy). A RobustScaler fit only on the training split is used for both features and target to reduce sensitivity to heavy-tailed moves.

PatchTST. We build on the original PatchTST implementation [6]. A channel-independent Transformer that segments each input channel into overlapping patches of length 8 with stride 4, projects each patch to a 128-dimensional latent vector, and processes the resulting patch sequence through three Transformer encoder blocks with eight attention heads. RevIN normalization is applied independently per channel and inverted at prediction time. Training uses AdamW [21] with cosine annealing and Huber loss [22] ($\delta = 0.003$).

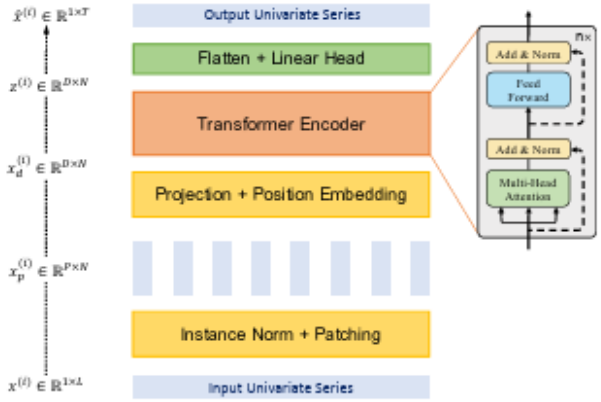

Figure 3: PatchTST architecture used for the patch-based time-series forecasting baseline.

TSMixer. We build on the original TSMixer implementation [7], which treats rows as timesteps and columns as input variates and alternates between time-mixing and feature-mixing MLP blocks. In the original design, the time-mixing MLP is shared across features, the feature-mixing MLP is shared across timesteps, and stacking N such mixer layers yields a parameter-efficient multivariate forecaster. Our implementation preserves this alternating mixer structure but introduces two task-specific modifications: an 8-dimensional learnable ticker embedding is appended to each sample to provide modest stock-specific specialization, and the original MSE objective is replaced with a combined Huber and quantile loss to improve robustness to heavy-tailed market moves.

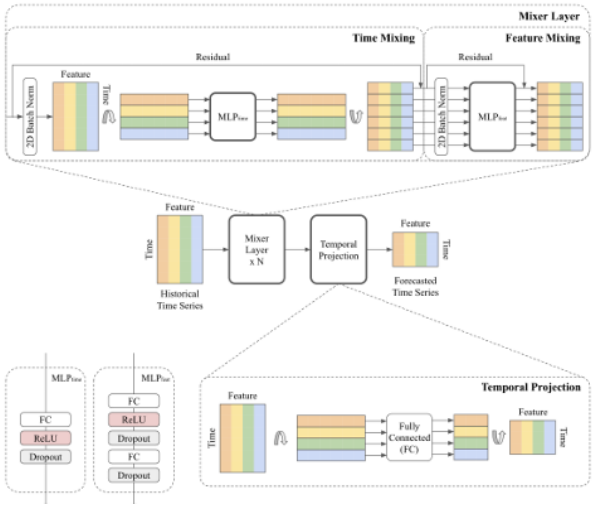

Figure 4: TSMixer architecture for multivariate time-series forecasting. Rows denote timesteps and columns denote input features. The model interleaves time-mixing and feature-mixing MLPs, with the number of mixer layers denoted by N. Adapted from [7].

Chronos Foundation Models. We evaluate two pre-trained time-series foundation models: Chronos T5-Large [2], an encoder-decoder Transformer pre-trained on large time-series corpora, and Chronos-2 [9]. Chronos T5-Large is evaluated across 842 S&P 500 tickers; Chronos-2 is evaluated on a 20% random sample of tickers (sampling ratio 0.2, seed 42) with the final 5 trading days reserved as ground truth and up to 2,048 prior business days used as context. Both models consume close-price sequences (transformed to log returns before inference, then mapped back to price space for evaluation) and produce probabilistic multi-step forecasts.

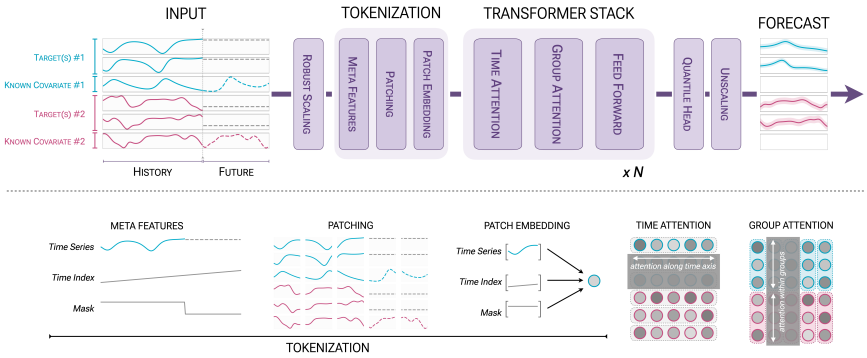

Figure 5: Chronos architecture diagram referenced in the unimodal temporal baseline discussion.

TCN-Transformer. A hybrid architecture in which four stacked temporal convolutional blocks with dilated causal convolutions (dilation factors 1, 2, 4, 8) extract local temporal features, which

are then treated as temporal tokens by a two-layer Transformer encoder with eight attention heads. Sinusoidal positional encodings are injected before the Transformer stage.

Temporal Fusion Transformer (TFT). Incorporates a Variable Selection Network (VSN) that learns per-variable importance weights using Gated Residual Networks (GRNs), an LSTM temporal encoder with gated skip connections, sinusoidal positional encoding, and interpretable multi-head attention with a shared value projection. The gated residual network design throughout the TFT allows it to function as a pass-through when any component's transformation is unhelpful, providing implicit regularization.

Both the TFT and TCN-Transformer baselines are trained on a 12-feature engineered representation (OHLCV plus Bollinger bands, RSI, and MACD components) with a 60-day lookback and a 5-day residual forecast horizon. The loss is Huber [22] with $\delta = 1.0$, features are standardized with a z-score scaler fit on the training split, the last time step of the encoder output is pooled before the linear prediction head, and the forecast increment is added to the last observed close price.

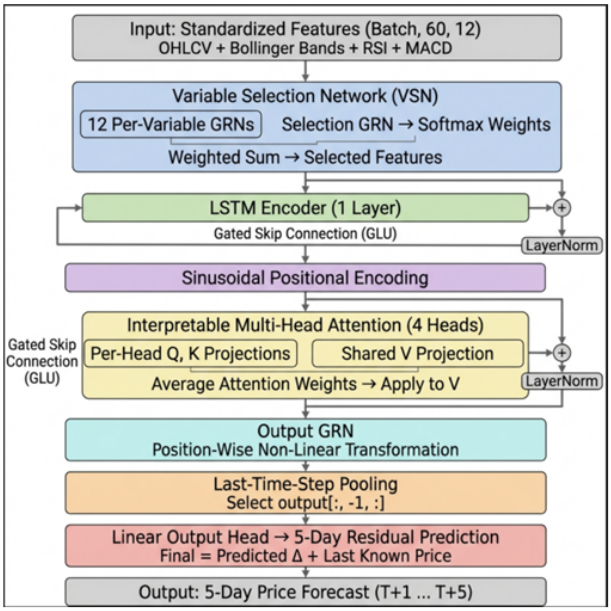

Figure 6: Temporal Fusion Transformer architecture used for the price-only baseline.

4.4.2 Temporal Encoder for Financial Time-Series (Chronos-T5 + FinBERT Fusion)

The ChronosFinBert model uses the encoder component of Chronos-T5-Base to encode a normalized close-price sequence of length $T \in \{30, 60, 90\}$ trading days. Across the ablation, $T = 60$ with log-return targets produced the best test RMSE, and all cross-model comparisons involving ChronosFinBert in Section 6 use that configuration. Scalar close prices are projected via a learned linear layer into the 512-dimensional encoder embedding space. Only the top four Transformer blocks are fine-tuned; the remaining encoder layers are frozen to preserve the temporal prior learned during pre-training. This produces a temporal representation matrix of shape $[B, T, 512]$.

Simultaneously, the most recent non-empty news headline is encoded by FinBERT, and the [CLS] token embedding (dimension 768) is extracted. The top four FinBERT encoder layers are fine-tuned. The temporal and sentiment embeddings are projected to a common fusion dimension of 512 via learned linear layers. A multi-head cross-attention mechanism is applied where the temporal embeddings serve as the query and the sentiment embedding, broadcast across the temporal axis, serves as key and value. This allows each temporal position to selectively attend to the sentiment context. The resulting fused representation is passed through a two-layer LSTM (hidden dimension 256, inter-layer dropout 0.2), and a linear forecast head projects the final hidden state to the 5-step prediction.

Figure 7: Overview of the ChronosFinBert multimodal forecasting architecture.

4.4.3 Multimodal Fusion Architectures (TFT/TCN + FinBERT)

Two multimodal fusion architectures are studied that pair the TFT or TCN-Transformer temporal encoder with frozen FinBERT. In both cases, each news article (up to 32 per sample) is independently encoded by frozen FinBERT to produce a [CLS] embedding. A four-head cross-article self-attention mechanism with a padding validity mask pools these per-article embeddings into a single 768-dimensional news context vector. The temporal encoder produces either a 128-dimensional (TFT) or 256-dimensional (TCN) context vector from the last timestep. Learned ticker embeddings (dimension 32) and sector embeddings (dimension 16) provide categorical context. All four representations are concatenated, passed through a LayerNorm, compressed to 256 dimensions through a dense bottleneck with ReLU and dropout, and projected to the five-step forecast via a linear head. The residual prediction strategy adds the forecast increment to the last observed close price.

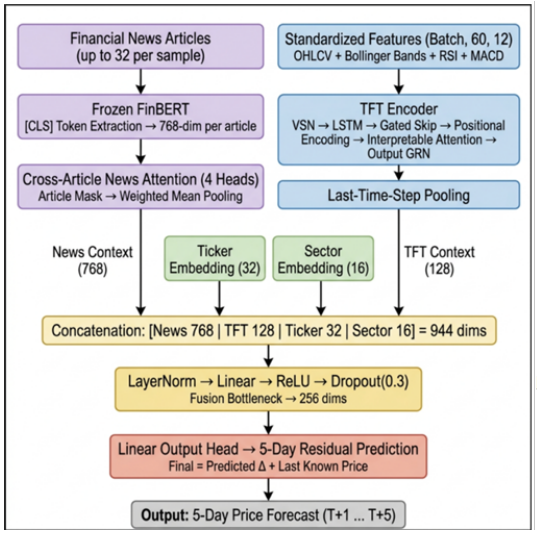

Figure 8: Late-fusion multimodal architecture combining TFT features with FinBERT news embeddings.

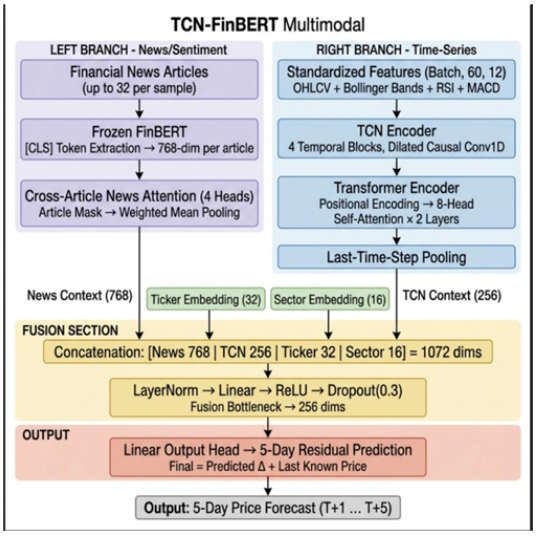

Figure 9: Late-fusion multimodal architecture combining TCN-Transformer features with FinBERT news embeddings.

4.4.4 FinBERT + CNN Price Encoder + Transformer Decoder

The most architecturally complete fusion model processes daily news articles through FinBERT with per-day article-level attention aggregation, producing a time-indexed sequence of daily news representations with positional encoding applied across the historical window. The full historical close price sequence is encoded by a dedicated one-dimensional convolutional encoder that captures local temporal structure through convolution, batch normalization, ReLU activation, and global average pooling. The resulting price representation and aggregated news representation are fused and used to condition a Transformer decoder. Learnable query tokens corresponding to each forecast step are combined with causal masking, ensuring each step is generated without access to future decoder positions. The decoder output is accumulated as price increments on top of the last observed value, producing the final forecast.

4.4.5 Graph Neural Network Architectures

GATv2 (Graph Attention Network v2). A late-fusion spatio-temporal architecture. Each stock's 20-day lookback is first encoded by a shared single-layer LSTM (hidden dimension 128) to produce a temporal node embedding. If FinBERT embeddings are enabled, the same-day 768-dimensional news vector is concatenated after temporal encoding, and the combined vector is projected to 256 dimensions via a linear layer with GELU activation and layer normalization. Two GATv2 convolution layers with four attention heads each perform relational message-passing, with the second layer producing a 64-dimensional bottleneck expanded back to 256 with a residual connection. A final MLP prediction head maps each node's representation to the scalar cumulative future log return. Training uses a composite objective combining Huber loss with a Pearson

cross-sectional ranking term, ensuring that the model is penalized both for magnitude error and for incorrect relative ordering of returns across stocks in the same day.

4.5 Fusion Strategies

Across the evaluated architectures, three distinct fusion paradigms emerge. Cross-attention fusion (ChronosFinBert) allows the temporal representation to selectively query the sentiment embedding at each timestep, producing a soft alignment between temporal positions and sentiment content. This is a fundamentally asymmetric design: the temporal signal drives the forecast, while sentiment provides context that modulates each temporal step. The result is that sentiment can be downweighted to near-zero at timesteps where it is uninformative without requiring an explicit gating signal.

Concatenation-based late fusion (TFT-FinBERT, TCN-FinBERT) independently computes temporal and sentiment embeddings, concatenates them, and passes the combined vector through a dense bottleneck. This design is simple and parameter-efficient but creates a modality imbalance problem: the 768-dimensional FinBERT embedding constitutes between 72% and 81% of the concatenated vector, potentially overwhelming the smaller temporal representation during optimization.

Hybrid encoder-decoder fusion (FinBERT+CNN+Transformer) integrates both modalities at multiple stages. The CNN encodes the full price history into a compact vector; FinBERT with per-day attention encodes the news stream with positional order preserved; these representations condition a Transformer decoder that generates each forecast step autoregressively. This design preserves the sequential structure of the forecast horizon and provides richer cross-modal conditioning than a single concatenation step.

5 Discussion

Three findings emerge from the benchmark that are worth separating carefully.

Fusion strategy matters more than the fact of being multimodal. Cross-attention fusion (ChronosFinBert) and encoder-decoder fusion (FinBERT+CNN+Transformer) produced competitive multimodal performance, while concatenation-based late fusion (TFT-FinBERT, TCN-FinBERT) degraded performance relative to the unimodal baselines built on the same temporal encoders. The direct cause is the modality imbalance already quantified in Section 4.5: the 768-dimensional FinBERT embedding occupies 72–81% of the fused vector, and once passed through a single bottleneck the temporal branch is effectively downweighted during optimization. Cross-attention and decoder conditioning avoid this by not forcing a shared bottleneck, letting the temporal signal drive the forecast while news provides context.

Evaluation spaces are not consistent across models. Models trained to predict log returns (PatchTST, TSMixer, the ChronosFinBert log-return variants, FinBERT+CNN+Transformer) report RMSE in the range 0.125–12.7896 in this benchmark. Models trained to predict standardized or dollar-level prices (TFT, TCN-Transformer, Chronos T5-Large, Chronos-2, concatenation-fusion variants) report RMSE in the range 0.0673–2.25. These numbers are not directly comparable: a small error in log-return space can map to a much larger error in price space once exponentiated, and vice versa. Any "best overall" claim within this benchmark should be qualified by the output space in which the model was evaluated. A cleaner future iteration of this benchmark should inverse-transform all predictions into a single unit — either daily log return or dollar price — before aggregating metrics. MAPE adds a second source of non-comparability even within the same output space, because its denominator becomes numerically unstable whenever targets approach zero, which is the common case for log returns.

Modalities contribute differently as a function of update frequency. News arrives at daily frequency and is closely aligned with short-horizon price reactions, which is why appropriately fused text helps. SEC fundamentals arrive quarterly and are forward-filled, so their information content is essentially step-wise on the daily trading calendar; at five- to seven-day horizons they are more useful as stabilizing firm-level context than as active predictors. Graph structure captures cross-sectional dependence rather than temporal state, and is therefore complementary to both sequence modeling and text fusion rather than substitutable for them.

Taken together, the results argue against framing the question as unimodal versus multimodal. The more informative framing is which fusion mechanism and which modality pairing is appropriate for a given horizon and target. For five- to seven-day forecasting of S&P 500 constituents, cross-attention fusion between a pretrained temporal encoder and a pretrained text encoder, and encoder-decoder fusion with an explicit price-sequence encoder, were the two multimodal configurations that delivered clean gains in our benchmark.

6 Results

Model	MSE	RMSE	MAE	MAPE (%)
Time Series				
LSTM	1.9723	1.4043	1.0880	319.53
PatchTST	3.04	2.36	1.44	—
TSMixer	162.45	12.7896	3.9467	—
Chronos-2	19.0126	1.3505	1.1954	—
Chronos T5-Large	—	1.3669	1.2130	**2.63**
TFT	**0.00452**	0.0673	0.0350	25.29
TCN-Transformer	0.00453	0.0673	0.0352	25.68
Time Series + News				
ChronosFinBert 60d	—	0.125	0.042	171.8
TCN-FinBERT	0.6592	0.8119	0.5217	177.14
TFT-FinBERT	1.3624	1.1672	0.7504	623.01
FinBERT+CNN+Transformer	—	1.45	0.93	—
FinBERT-TCN Multimodal	—	2.25	1.53	—
Time Series + News + Sectors				
GAT	—	—	—	—

Table 2: Comparative performance summary across unimodal, multimodal, and graph-based forecasting models. Best values in each metric column are shown in bold, with missing entries excluded from the comparison.

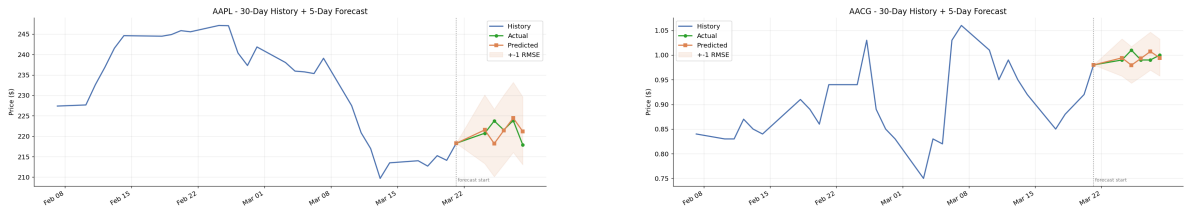

Figure 10: LSTM forecasts on example tickers.

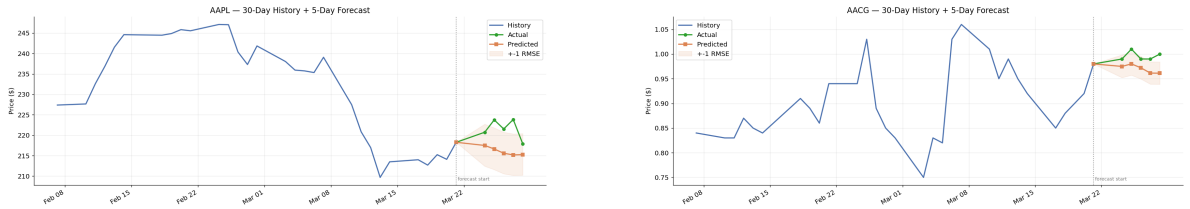

Figure 11: PatchTST forecasts on example tickers.

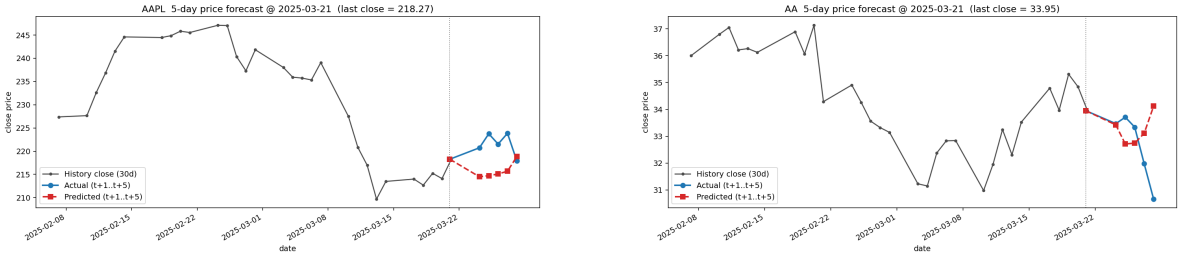

Figure 12: TSMixer forecasts on example tickers.

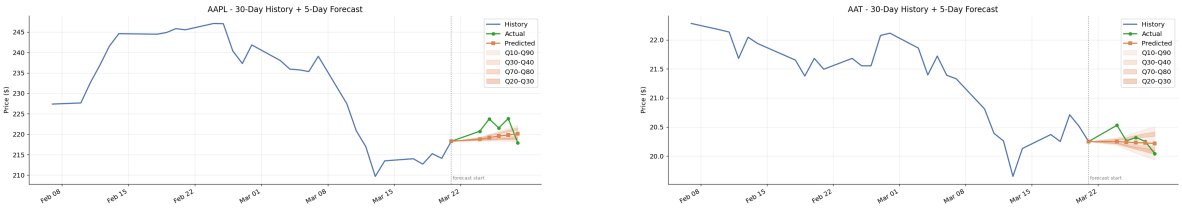

Figure 13: Chronos T5-Large forecasts on example tickers.

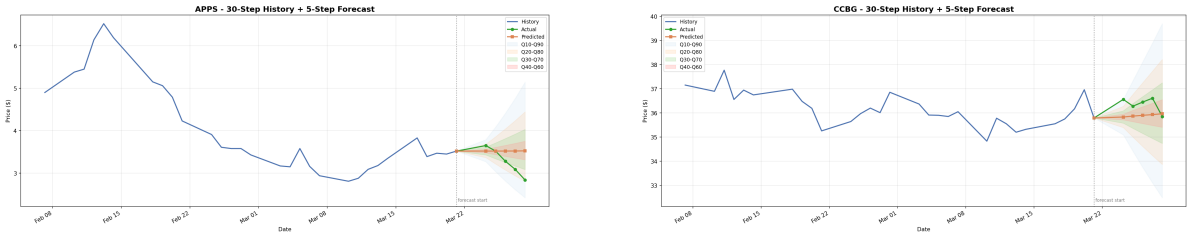

Figure 14: Chronos-2 forecasts on example tickers.

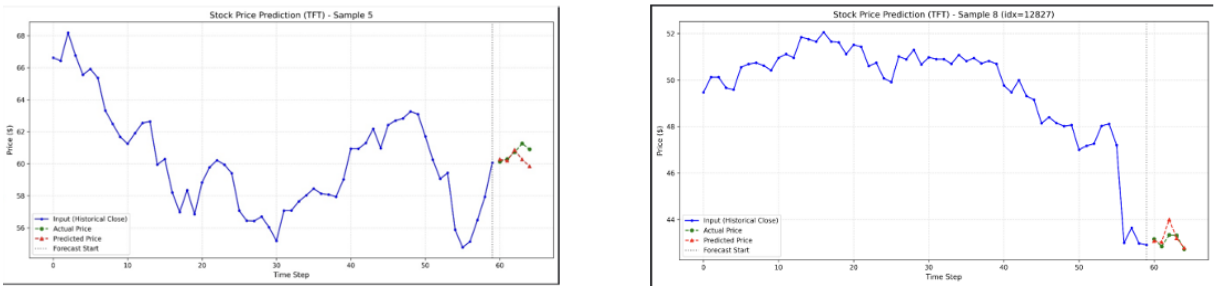

Figure 15: TFT price forecasting results on representative stocks.

7 Conclusion

This study benchmarked unimodal temporal encoders, pretrained time-series foundation models, multimodal fusion architectures, and graph neural networks on the S&P 500 subset of the FinMultiTime dataset, using short-horizon close-price and log-return forecasting as the common task.

The strongest standalone point-forecast performance came from TFT among the purely temporal encoders in the reported benchmark, while Chronos T5-Large achieved the lowest reported MAPE at 2.63%, an sMAPE of 2.61%, and the highest zero-shot directional accuracy at 62.9% median. These results suggest useful transfer from time-series pretraining to equity returns even without fine-tuning.

Among fusion architectures, the choice of fusion operator dominated the choice of temporal encoder. Cross-attention fusion and encoder-decoder fusion improved on their unimodal baselines; concatenation-based late fusion did not, because of the modality-imbalance effect quantified earlier. Graph neural networks with GATv2 convolution layers and a combined Huber-plus-ranking loss are a reasonable default when cross-sectional structure matters.

Several limitations should temper these conclusions. Evaluation metrics are not yet on a common output scale across architectures, which complicates cross-model ranking. The news modality is aggregated in different ways by different models — most-recent headline, daily attention pool, or cross-article attention — and we did not isolate the effect of aggregation strategy from the effect of fusion strategy. Five- to seven-day horizons do not test whether fundamentals or graph structure become more important at longer horizons.

Future work should standardize the evaluation space, extend the benchmark to longer horizons where fundamentals should carry more weight, add gated or mixture-of-experts fusion to address modality imbalance directly, and move evaluation from point-forecast error to portfolio-level metrics that reflect downstream decision value.

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